Thus, only vortices corresponding to the minus sign can be formed; the vortex moves in a direction opposite to the main fluid flow.

The condition (8.40) is most easily satisfied when the vortex radius r is large. The latter, however, cannot exceed the pipe radius R. The first instability should thus appear when v_s reaches the "critical" value

$$v_c = \frac{\hbar}{mR} \log \frac{R}{\xi} \tag{8.41}$$

Such a derivation of the critical velocity is admittedly very crude, since it relies on a model which ignores many relevant features of the problem (such as, for instance, the image effect). Nevertheless, (8.41) is in fair agreement with the experimental data on helium; the observed dependence of v_c on the pipe radius R is of the predicted form. Actually, the success of (8.41) is mostly of a dimensional nature. After all, there are not so many parameters in the problem: any theory based on vortex formation is bound to give a critical velocity similar to (8.41). Therefore, the agreement of (8.41) with experiment does not substantiate our simple model (which is probably quite unphysical); it only proves that superfluid flow is limited by *some* vortex instability. The purpose of the above calculation was to provide an order of magnitude of v_c, not to describe what really happens.

CHAPTER 9

MICROSCOPIC THEORY: UNIFORM CONDENSATE[1]

Thus far we have been concerned primarily with macroscopic properties of superfluid Bose liquids in general, and of He II in particular. Where we have been concerned with microscopic properties, as in Chapter 3, we have discussed phenomenological theories of superfluid behavior. We now wish to turn our attention to microscopic theories of interacting Bose systems, that class of theories which begin with a given law of interaction between the particles, and derive the various properties of the system therefrom.

Microscopic theory does not, at present, provide a quantitative description of liquid He II. Indeed, there exists a satisfactory microscopic theory of an interacting boson system in only two limiting cases, those of

1. Weak repulsive interactions between the bosons.
2. A dilute gas with arbitrary repulsive interaction between the particles.

Nonetheless, a discussion of the microscopic theory of superfluid boson systems is of considerable interest. One sees, quite generally, how the macroscopic occupation of a single quantum state dictates the structure of the microscopic description of system behavior. Moreover, the above limiting cases provide explicit examples of "model" systems which behave in accord with the general principles we have set forth in the preceding chapters. In the present chapter we shall consider the equilibrium behavior of systems at $T = 0$, which, further, possess a uniform condensate;

[1]The treatment in this chapter follows closely that in Hugenholtz and Pines (1959) and Pines (1963).

in the following chapter, we consider a spatially-varying condensate. Our primary emphasis will be on the derivation of the elementary excitation spectrum and response functions for the model systems mentioned above. We shall not discuss microscopic theories of superfluid behavior at finite temperatures or under non-equilibrium conditions.

9.1 Basic Hamiltonian

The Hamiltonian for our system of interacting bosons may be written in the form:

$$H = \sum_{\mathbf{p}} \varepsilon_p^o a_{\mathbf{p}}^+ a_{\mathbf{p}} + \frac{1}{2} \sum_{\mathbf{pkq}} V_q a_{\mathbf{p+q}}^+ a_{\mathbf{k-q}}^+ a_{\mathbf{k}}\, a_{\mathbf{p}} \tag{9.1}$$

where $\varepsilon_p^o = p^2/2m$ is the free particle kinetic energy, and V_q is the Fourier-transform of the interaction potential between a pair of particles. The Hamiltonian (9.1) serves as a starting Hamiltonian in our consideration of microscopic theory; actually, we shall show that the existence of macroscopic occupation of a single quantum state leads to an effective Hamiltonian which looks rather different.

We assume that the particles are condensed in the state with zero momentum, and that the number of particles in that state is

$$N_o \gg 1 \tag{9.2}$$

As we have discussed in Chapter 2, the condensate is depleted as a result of the interaction between the particles; consequently N_o may be only a small fraction of N. Nonetheless, as long as that fraction is independent of the size of the system, the condition (9.2) is obviously well satisfied for a system of macroscopic size.

Because N_o is so large, one would expect that the addition or subtraction of a few particles in the zero-momentum state would not have any appreciable effect on the overall physical behavior of the system. Such a statement appears innocuous, as well as being nearly obvious. In fact, it has profound consequences for the development of a microscopic theory, as Bogoliubov first showed in 1947. To see how this comes about, we consider the behavior of the annihilation and creation operators for the zero-momentum state. When acting on the ground state wave-function, ψ_o, they yield:

$$a_o|\psi_o(N_o)\rangle = \sqrt{N_o}|\psi_o(N_o - 1)\rangle \tag{9.3}$$

$$a_o^+|\psi_o(N_o)\rangle = \sqrt{N_o + 1}|\psi_o(N_o + 1)\rangle \tag{9.4}$$

where $\psi_o(N_o)$ is the ground state wave-function with N_o particles in the condensate, $\psi_o(N_o-1)$ that with N_o-1 particles there, etc. Since $N_o \gg 1$, one may do two things:

1. Replace $\sqrt{N_o+1}$ by $\sqrt{N_o}$.
2. Assume that the difference between $\psi_o(N_o - 1)$, $\psi_o(N_o)$, and $\psi_o(N_o+1)$ is of order $1/N$; that is, removing or adding a particle in the zero-momentum state *in no way* alters the physical properties of the system.

Under these circumstances, a_o and a_o^+ will commute with each other; since they already commute with all $a_{\mathbf{p}}$ and $a_{\mathbf{p}}^+$ for $\mathbf{p} \neq 0$, they may be replaced everywhere by the "c" number, $\sqrt{N_o}$. This is the famous Bogoliubov prescription; it amounts to treating the condensate as a static quantity and so neglecting any possible behavior it might have as a consequence of its coupling to an excited particle. This is clearly an excellent approximation: the *mass* of the condensate being of order N times the mass of any single particle, dynamic effects in particle-condensate interaction can scarcely be expected to play any role.

On applying the Bogoliubov prescription, one obtains the following Hamiltonian:

$$\begin{aligned} H = &\sum_{\mathbf{p}}{}' \left(\varepsilon_p^o + N_o\, V_o \;+ N_o\, V_{\mathbf{p}}\right) a_{\mathbf{p}}^+ a_{\mathbf{p}} \\ &+ \sum_{\mathbf{p}}{}' \frac{N_o V_p}{2} \left(a_{\mathbf{p}}^+ a_{-\mathbf{p}}^+ + a_{\mathbf{p}}\, a_{-\mathbf{p}}\right) + \frac{1}{2} N_o^2 V_o \\ &+ \sum_{\mathbf{pq}}{}' \sqrt{N_o} \frac{V_q}{2} \left(a_{\mathbf{p+q}}^+ a_{\mathbf{q}}\, a_{\mathbf{p}} + a_{\mathbf{p+q}}^+ a_{-\mathbf{q}}^+ a_{\mathbf{p}}\right) \\ &+ \sum_{\mathbf{pkq}}{}' \frac{V_q}{2} a_{\mathbf{p+q}}^+ a_{\mathbf{k-q}}^+ a_{\mathbf{k}}\, a_{\mathbf{p}} \end{aligned} \tag{9.5}$$

The primes are intended to remind the reader that nowhere do the creation and annihilation operators for the zero-momentum state appear. The zero-momentum state has disappeared from the problem, its only residue being the various factors of N_o which appear in (9.5). We see from the interaction terms in (9.5) that the number of bosons is not conserved.

There appear terms which create or destroy two bosons, scatter one into two, and vice-versa. Put another way, the particle number operator,

$$N'_{op} = \sum_{\mathbf{p}}{}' a^+_{\mathbf{p}} a_{\mathbf{p}} \tag{9.6}$$

which represents the number of particles which are *not* in the condensate, does not commute with the Hamiltonian, (9.5). We are taught in quantum mechanics that particle conservation is a Good Thing; to what extent, then, can one preserve it, while still making the quite physical approximation of treating the condensate as a fixed quantity?

Our original problem was to determine the ground state of the system of N interacting bosons, that is, the eigenstate of the Hamiltonian, (9.1), which has the ground state energy, E_o, subject to the condition that the number of particles is equal to N. In the modified problem, specified by the Hamiltonian (9.5), we must therefore impose the subsidiary condition

$$\langle N'_{op} \rangle = N'(N_o) = N - N_o \tag{9.7}$$

N_o then appears as a parameter, which is to be determined in such a way that the system energy is minimized [subject to the constraint (9.7)].

If N' commuted with H, one could satisfy the subsidiary condition (9.7), by imposing it on the unperturbed system wave-function. It would then automatically be satisfied for the true wave-function of the system. Such a possibility being not available, Hugenholtz and Pines (1959) have put forth an alternative method, based on a Lagrangian multiplier technique. They proposed that in place of (9.5) one consider the Hamiltonian,

$$H' = H(N_o) - \mu N'_{op} \tag{9.8}$$

while relaxing the subsidiary condition (9.7). The ground state wave-function appropriate to (9.8) is then a function of N_o and μ, $\psi'_o(N_o, \mu)$. The various expectation values of interest are likewise functions of both N_o and μ. The parameter μ, which turns out to be the chemical potential, is determined by the condition

$$N'(N_o, \mu) + N_o = N \tag{9.9}$$

where $N'(N_o, \mu)$ is the expectation value,

$$\langle \psi'_o(N_o, \mu) | N'_{op} | \psi'_o(N_o, \mu) \rangle = N'(N_o, \mu) \tag{9.10}$$

With such a choice of μ, $\psi'_o(N_o, \mu)$ is the ground state of (9.8) with the subsidiary condition (9.10). The remaining parameter, N_o, is determined by the condition that for fixed N,

$$\frac{dE_o}{dN_o} = 0 \tag{9.11}$$

A little algebra (cf. Hugenholtz and Pines) leads one to the following equivalent expression for μ,

$$\mu = \frac{\partial E'_o}{\partial N_o} = \frac{\partial E_o}{\partial N} \tag{9.12}$$

which shows that μ is indeed the chemical potential. The replacement of H by H', (9.8), is thus seen to be equivalent to replacing the particle kinetic energy, $p^2/2m$, by the value $p^2/2m - \mu$, that is, measuring the single particle energies from the chemical potential μ [which is given, in turn, by (9.12)].

The procedure we have described is needed whenever there is appreciable depletion of the zero-momentum state as a result of particle interaction. Such will certainly be the case for He II [we recall the estimate of a 92% depletion by Penrose and Onsager (1956)]. One need not, in fact, follow this procedure for the examples we treat in this section, since in both cases the depletion is quite small, and N_o may be simply replaced by N, as in the original work of Bogoliubov. We have gone into the above procedure in some detail because it illustrates clearly the extent to which particle-conservation must be abandoned in constructing a microscopic theory of superfluid behavior. We see that it is possible to maintain particle conservation on the average, whilst using a non-particle conserving Hamiltonian. Any departure from particle conservation is then unimportant, and must be ascribed to fluctuations in the condensate population.

The appearance of non-particle-conserving states is perhaps obvious in the case of a superfluid boson system. They are equally important for a fermion superfluid. We may say that superfluidity (boson *and* fermion) is characterized by macroscopic occupation of a single quantum state, and that such state occupation and lack of particle conservation go hand in hand. It is possible to construct, in principle, theories of superfluid behavior in which such states do not appear. In practice the resulting theories are clumsy and complicated and yield results in agreement with the non-conserving theory we consider here.

9.2 Weak Coupling Approximation

We consider now the solution of the Hamiltonian in a weak coupling approximation, due to Bogoliubov (1947). It is based on the following intuitive idea. In the ground state of the non-interacting system, all particles have zero momentum. Hence, when one takes into account interactions, those which are most important are those which involve the maximum number of zero-momentum particles. One can classify the interaction terms in $H(N_o)$ according to the powers of $\sqrt{N_o}$ they contain. On this basis, one is led to divide H', (9.8), into two parts,

$$H' = H_1 + H_2 \tag{9.13}$$

where

$$H_1 = \sum_{\mathbf{q}}{}' \tilde{\varepsilon}_{\mathbf{q}} a^+_{\mathbf{q}} a_{\mathbf{q}} + \sum_{\mathbf{q}}{}' N_o \frac{V_q}{2} \left(a^+_{\mathbf{q}} a^+_{-\mathbf{q}} + a_{\mathbf{q}} a_{-\mathbf{q}}\right) + \frac{1}{2} N_o^2 V_o \tag{9.14}$$

$$H_2 = N_o \sum_{\mathbf{pq}}{}' \frac{V_q}{2} \left(a^+_{\mathbf{p+q}} a_{\mathbf{q}} a_{\mathbf{p}} + a^+_{\mathbf{p+q}} a^+_{-\mathbf{q}} a_{\mathbf{p}}\right)$$

$$+ \sum_{\mathbf{pkq}}{}' \frac{V_q}{2} a^+_{\mathbf{p+q}} a^+_{\mathbf{k-q}} a_{\mathbf{k}} a_{\mathbf{p}} \tag{9.15}$$

and

$$\tilde{\varepsilon}_q = \varepsilon^o_q + N_o V_o + N_o V_q - \mu \tag{9.16}$$

H_1 is assumed to contain the most important parts of the particle interaction. H_2 gives rise to correction terms, which are assumed negligible in lowest order of approximation.

The Hamiltonian, H_1, may be diagonalized by means of a transformation to new variables, $\alpha_{\mathbf{p}}$ and $\alpha^+_{\mathbf{p}}$:

$$\begin{aligned} a_{\mathbf{p}} &= u_p \alpha_{\mathbf{p}} - v_p \alpha^+_{-\mathbf{p}} \\ a^+_{\mathbf{p}} &= u_p \alpha^+_{\mathbf{p}} - v_p \alpha_{-\mathbf{p}} \end{aligned} \tag{9.17}$$

If u_p and v_p are real and satisfy the conditions

$$u_p^2 - v_p^2 = 1 \tag{9.18}$$

the transformation is readily seen to be canonical; the new creation and annihilation operators, $\alpha^+_{\mathbf{p}}$ and $\alpha_{\mathbf{p}}$, thus obey the usual boson commutation rules:

$$\left[\alpha_{\mathbf{p}}, \alpha^+_{\mathbf{p}'}\right] = \delta_{\mathbf{p},\mathbf{p}'}$$

$$\left[\alpha_{\mathbf{p}}, \alpha_{\mathbf{p}'}\right] = \left[\alpha^+_{\mathbf{p}}, \alpha^+_{\mathbf{p}'}\right] = 0 \tag{9.19}$$

The reader may readily verify that in the new representation the off-diagonal terms in H are eliminated if one takes:

$$u_q^2 = \frac{1}{2}\left[1 + \frac{\tilde{\varepsilon}_q}{\omega_q}\right] \tag{9.20}$$

$$v_q^2 = \frac{1}{2}\left[\frac{\tilde{\varepsilon}_q}{\omega_q} - 1\right] \tag{9.21}$$

where

$$\omega_q^2 = \tilde{\varepsilon}_q^2 - N_o^2 V_q^2 \tag{9.22}$$

H_1 then takes the following form:

$$H_1 = \sum_{\mathbf{q}}{}' \left\{\omega_q \alpha^+_{\mathbf{q}} \alpha_{\mathbf{q}} + \frac{1}{2}\left(\omega_q - \tilde{\varepsilon}_q\right)\right\} + \frac{1}{2} N_o^2 V_o \tag{9.23}$$

derive these results

The Bogoliubov transformation leads to a new set of *independent* quasi-particles, which are characterized by the creation operators, $\alpha^+_{\mathbf{q}}$, and energy ω_q. The ground state of the system is defined as the "vacuum" of such quasi-particles, and is thus specified by the relation

$$\alpha_{\mathbf{q}}|0\rangle = 0 \tag{9.24}$$

In order to completely determine ω_q, we must calculate the chemical potential μ. This is easily done to lowest order in V_q: on making use of (9.12) and noting that the leading term in the ground state energy is $1/2\left(N_o^2 V_o\right)$, we find

$$\mu = N_o V_o \tag{9.25}$$

Hence

$$\tilde{\varepsilon}_q = \frac{q^2}{2m} + N_o V_q \tag{9.26}$$

while

$$\omega_q = \left(\frac{q^2 N_o V_q}{m} + \frac{q^4}{4m^2}\right)^{1/2} \tag{9.27}$$

Since V_q approaches a constant, V_o, for small q, the new quasi-particles are seen to be phonons in the long wave-length limit, with a velocity

$$s_B = \left(\frac{N_o V_o}{m}\right)^{1/2} \tag{9.28}$$

At short wave-lengths the quasi-particles behave like almost free particles, with an energy

$$\varepsilon_{\rm HF} = \frac{q^2}{2m} + N_o V_q = \tilde{\varepsilon}_q \tag{9.29}$$

This latter energy is just the energy a quasi-particle would have in the Hartree-Fock approximation, since, in this approximation the second term of H_1 may be neglected altogether.

We see that the dramatic new features of the boson system appear at long wave-lengths, specifically for

$$\lambda \gg \lambda_c = \left(\frac{1}{2ms_B}\right) \tag{9.30}$$

λ_c is the characteristic length h for the transition from phonon-like behavior to single particle-like behavior. The fact that in a consistent microscopic theory the long wave-length quasi-particles would be phonons was certainly to be expected on the basis of our derivation in Chapter 2. The nature of the quasi-particles in this limit could not, however, have been anticipated. On inverting the transformation, (9.17), we find

$$\alpha_{\mathbf{q}}^+ = u_q\, a_{\mathbf{q}}^+ + v_q\, a_{-\mathbf{q}} \tag{9.31}$$

while

$$\lim_{q\to 0} u_q = v_q = \sqrt{\frac{ms_B}{2q}} \tag{9.32}$$

Hence a long wave-length quasi-particle of momentum q is nearly in equal parts a "bare" particle of momentum q and a bare "hole" of momentum $-q$.

Such behavior contrasts sharply with that of a normal fermion system. There we saw that a quasi-particle consists in configurations involving a bare particle, two bare particles and one hole, etc., all of which have the same particle number. The present situation reflects clearly the lack of particle conservation. The interaction term in H_1 which is responsible is that which creates or destroys two bare particles; by means of it, a particle of momentum q may be transformed, in part, into a hole of momentum $-q$. At small q this term dominates the behavior of the system particles to such an extent that the resulting quasi-particle is equally shared between a bare particle and a bare hole. This aspect of the Bogoliubov theory is common to all existing microscopic theories. It shows clearly how the condensate acts as a reservoir (or sink) of particles.

The depletion of the condensate as a result of particle interaction is readily calculated. One finds:

$$N - N_o = \langle 0|\sum_{\mathbf{p}}{}' a_{\mathbf{p}}^+ a_{\mathbf{p}}\,|0\rangle = \sum_{\mathbf{p}} v_p^2 = \frac{1}{2}\sum_{\mathbf{p}}\left\{\frac{\varepsilon_p + N_o V_p}{\omega_p} - 1\right\} \tag{9.33}$$

It is seen to be of order V_p, and hence is negligible to the extent that the interaction is weak. We may, therefore, replace N_o by N everywhere it appears in the foregoing (apart from the lowest-order Hartree term in E_o).

It is left as an exercise to the reader to show that the dynamic form factor is given by:

$$s_B(\mathbf{q},\omega) = \frac{N_o q^2}{2m\omega_q}\delta\left(\omega - \omega_q\right) \simeq \frac{Nq^2}{2m\omega_q}\delta\left(\omega - \omega_q\right) \tag{9.34}$$

One verifies readily that $s_B(\mathbf{q},\omega)$ satisfies the sum rules, (2.20) and (2.21) [in computing the right-hand side of the latter, one works only to lowest order in μ; that is μ is given by (9.25)]. The density-density response function is:

$$\chi_B(\mathbf{q},\omega) = \frac{Nq^2/m}{\omega^2 - \omega_q^2} \tag{9.35}$$

We see that for all values of $\mathbf{q}$, one has only single quasi-particle excitations contributing to both $S(\mathbf{q},\omega)$ and $\chi(\mathbf{q},\omega)$, a feature which is special to the Bogoliubov approximation. There are no contributions due to multi-particle excitations from the condensate, because such arise only via H_2, which we have neglected. The correlation length defined via (9.30) is therefore not obviously the same as either of the coherence lengths we have introduced in Chapter 4; in fact, closer investigation shows that it is identical to both for the case of a pure system.

We have asserted that the Bogoliubov approximation of keeping only H_1 is a weak coupling approximation. It is not that in the normal sense in which, say, one keeps only the second-order term in the particle interaction. Rather it is the lowest-order *consistent* weak-coupling approximation and is, in fact, equivalent to the random phase approximation. We now develop this thesis in some detail.

From the standpoint of perturbation theory [Brueckner and Sawada (1957)], the lowest order calculation is the Hartree-Fock approximation. In this, the ground state wave function is

$$\psi_{\rm HF} = (a_o^+)^{N_o}|0\rangle \tag{9.36}$$